9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

−5

0

5

10

15

− 10 −5 0 5 10

UMAP_1

UMAP_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

16
17
18
19
20
21
22
23
24
25
26
27
28
29
30

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

16
17
18
19
20
21
22
23
24
25
26
27
28
29
30

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

P2RY12

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

P2RY12

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CX3CR1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CX3CR1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

TREM2

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

TREM2

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

APOE

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

APOE

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

FTL

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

FTL

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

CD83

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

CD83

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

NFKB1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

NFKB1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD2

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD2

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

SKAP1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SKAP1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

CD14

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

CD14

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD68

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD68

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

GFAP

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

GFAP

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

AQP4

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

AQP4

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

SLC1A2

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

SLC1A2

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

SLC4A4

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SLC4A4

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

HSP90AA1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

HSP90AA1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

SERPINH1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

SERPINH1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

DNAH9

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

DNAH9

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

SPAG17

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SPAG17

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

VIM

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

VIM

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

S100B

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

S100B

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

IL1RAPL1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

IL1RAPL1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

KIRREL3

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

KIRREL3

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

MRC1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

MRC1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD163

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD163

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD163L1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD163L1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

MSR1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

MSR1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

CLDN5

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

CLDN5

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD38

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD38

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

CD79A

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

CD79A

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

IGHM

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

IGHM

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

IGHG1

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

IGHG1

9

6

1

3 18

0 12

17
2

4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 10

0

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

TSPO

9

6

1

3

18

0

12

17

2
4

5

21

11

23

14

28

15

8

22

20

26

7

10

16

25

19

30

27

13

29

24

− 50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

TSPO